BRAC UNIVERSITY
Inspiring Excellence

CSE340: Computer Architecture
Spring 2026
Quiz - 04
Duration: 25 minutes

A

Name:	ID:	Section:

1. Given, cache size = 4 blocks and main memory block size = 16 bits.

a. Simulate the given cache table for each of the memory addresses accessed by the CPU using a direct-mapped cache. For each address, you must show the **index, tag, and offset** bits in your workings. Mention whether each access is a **hit or miss.** [10]

Current State of the Cache Table:

Index	Offset	Valid	Tag	Data
00	0	N		
	1	~~N~~ Y	~~010~~ ~~110~~ 010	~~Data~~ ~~Data~~ data
01	0	Y	001	Data
	1	N		
10	0	Y	~~010~~ 110	~~Data~~ Data
	1	Y	010	Data
11	0	~~N~~ Y	011	data
	1	N		

Access History:

Memory Address in decimal	Cache Hit/Miss
10	hit
30	miss
17	miss
49	miss
52	miss
21	hit
30	hit
17	miss

b. What is the Hit ratio for the given access history [1]
c. What is the miss ratio for the given access history? [1]

Question 2: True or False [3]

1. In a direct-mapped cache, each memory block can be placed in exactly one cache location. True
2. Temporal locality refers to the tendency to access memory locations that are close together in address space. False
3. SRAM is slower but cheaper per bit than DRAM. False

Solution:

Index size = log2(4) = 2 bits, Offset Size = log2(2) = 1 bit, Tag size = 6-2-1 = 3

10 = 001010, Tag = 001, Index = 01, Offset = 0

30= 011110, Tag = 011, Index = 11, Offset = 0

17= 010001, Tag = 010, Index = 00, Offset = 1

49= 110001, Tag = 110, Index = 00, Offset = 1

52= 110100, Tag = 110, Index = 10, Offset = 0

21= 010101, Tag = 010, Index = 10, Offset = 1

30= 011110, Tag = 011, Index = 11, Offset = 0

17= 010001, Tag = 010, Index = 00, Offset = 1

Hit Ratio = ⅜

Miss Ratio = ⅝

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Name:	ID:	Section:

1. Given, cache size = 4 blocks and main memory block size = 16 bits.

a. Simulate the given cache table for each of the memory addresses accessed by the CPU using a direct-mapped cache. For each address, you must show the **index, tag, and offset** bits in your workings. Mention whether each access is a **hit or miss.** [10]

Current State of the Cache Table:

Index	Offset	Valid	Tag	Data
00	0	N		
	1	Y	011	Data
01	0	Y	110	Data
	1	~~N~~ Y	~~010~~ ~~101~~ 010	~~Data~~ ~~Data~~ Data
10	0	N		
	1	~~N~~ Y	100	Data
11	0	N		
	1	Y	~~101~~ 001	Data data

Access History:

Memory Address in decimal	Cache Hit/Miss
15	miss
19	miss
25	hit
37	miss
15	hit
43	miss
19	miss
50	hit

b. What is the Hit ratio for the given access history [1]

c. What is the miss ratio for the given access history? [1]

Question 2: True or False [3]

1. Spatial Locality refers to the tendency to access the same information frequently. False
2. SRAM is typically used for cache memory due to its faster access time. True
3. In a direct-mapped cache, the miss penalty is usually very low. False

Solution:

Index size = log2(4) = 2 bits, Offset Size = log2(2) = 1 bit, Tag size = 6-2-1 = 3

15 = 001111, Tag = 001, Index = 11, Offset = 1

19= 010011, Tag = 010, Index = 01, Offset = 1

25= 011001, Tag = 011, Index = 00, Offset = 1

37= 100101, Tag = 100, Index = 10, Offset = 1

15= 001111, Tag = 001, Index = 11, Offset = 1

43= 101011, Tag = 101, Index = 01, Offset = 1

19= 010011, Tag = 010, Index = 01, Offset = 1

50= 110010, Tag = 110, Index = 01, Offset = 0

Hit Ratio = ⅜

Miss Ratio = ⅝